A Weighted Stationarity View of Two-Layer Feature Learning

Angelo Farfan
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Abstract

The Neural Feature Ansatz (NFA) observes that a trained network's first-layer feature matrix tracks an average gradient outer product (AGOP). Starting from the first-layer stationarity equation in a two-layer model with square loss and L_2 regularization on the first layer, I find that the object most directly controlled at convergence is not raw AGOP but a *residual-weighted* AGOP, $H^2 \approx \kappa_{\mathrm{eff}}\,\widetilde{G}$, with $\widetilde{G}$ the uncentered second moment of the input gradients of the loss. Raw AGOP is a derived, regime-dependent shadow of this law, gated by two checkable links: a scalar *beta bridge* and *high-gain pair closure*. The weighted-law error splits exactly into a pushed pair-isotropy term and a stationarity defect, and the pushed term, not the global pair defect, is the operative one: a large global defect can coexist with a tiny pushed one. I isolate the failure mechanisms, give finite-dimensional examples showing the two links are logically independent, and test the diagnostics across nine synthetic teacher-student families, where the weighted law proves robust even to deliberate pair-closure attacks and degrades only under persistent-residual outlier structure, which also bends the beta link.

1. Introduction

The NFA, posited by Radhakrishnan et al. [**?**] and central to recursive feature machines, states that the learned first-layer geometry $H := B^\top B$ of a trained network is proportional to a power of the AGOP $G := \frac{1}{n}\sum_i \nabla f(x_i)\nabla f(x_i)^\top$. Empirically the relation is striking, but a general nonlinear theory is open: for deep networks, NFA-type statements have been proved only in linear regimes, with nonlinear counterexamples [**?**]. The Features-at-Convergence viewpoint (FACT) [**?**] reframes the question, deriving a self-consistent feature relation from first-order optimality rather than fitting AGOP after training, and explaining when NFA holds or fails.

This report takes that viewpoint literally in the simplest nonlinear model that has the structure: a two-layer network trained with square loss and ridge on the first layer. The stationarity equation suggests a slightly different picture, which I organize around three observations. **(1) The natural object is residual-weighted, not raw**: stationarity yields a deterministic identity $H^2 - \kappa_{\mathrm{eff}}\widetilde{G} = (\lambda^2 n^2)^{-1} B^\top (S - d_{\mathrm{eff}} T)B + B^\top E_{\mathrm{stat}} B$, where $\widetilde{G}$ is the residual-weighted AGOP, which equals the uncentered second moment of the input gradients of the loss, so the convergence-side object is derived rather than assumed. **(2) Raw AGOP needs two extra links**: a scalar *beta bridge* ($\widetilde{M} \approx \beta M$) converts $\widetilde{G}$ into G, and the weighted law itself requires *high-gain pair closure*. I give exact identities for both and the failure taxonomy they imply, and a 2-dimensional argument shows the two links are logically independent, so no theorem with hypotheses limited to stationarity and finite-dimensional algebra can imply a universal raw NFA law. **(3) The right pair diagnostic is not the global one**: the pushed pair error $B^\top(S - d_{\mathrm{eff}}T)B$ factors through a stationarity-induced gain map, so a large global pair-isotropy defect can be conservative. The contribution is this finite-sample decomposition, the diagnostics that read those links off any checkpoint, and a synthetic-network study mapping which link fails where. Closely related concurrent work studies $B^\top B$ as a feature-dynamics object via a "Virtual Covariance" [**?**], and feature-learning *dynamics* in two-layer networks from small initialization is studied separately [**?**].

2. Setup and first-order identities

The model is $f(x) = a^\top \phi(Bx)$ with $B \in \mathbb{R}^{m\times d}$, $a \in \mathbb{R}^m$, smooth activation ϕ. With training data $(x_i, y_i)_{i=1}^n$, residuals $r_i := f(x_i) - y_i$, and hidden-derivative features $q_i := a \odot \phi'(Bx_i)$, write $X = [x_1, \ldots, x_n]$, $Q = [q_1, \ldots, q_n]$, $R = \mathrm{diag}(r_1, \ldots, r_n)$. By the chain rule $\nabla f(x_i) = B^\top q_i$. Define

$$M := \tfrac{1}{n}QQ^\top, \qquad \widetilde{M} := \tfrac{1}{n}QR^2Q^\top, \qquad G := B^\top M B, \qquad \widetilde{G} := B^\top \widetilde{M} B = \frac{1}{n}\sum_i r_i^2 \nabla f(x_i)\nabla f(x_i)^\top.$$

Since the per-sample loss is $\ell_i = \frac{1}{2}r_i^2$ and $\nabla_x \ell_i = r_i \nabla f(x_i)$, the weighted object is exactly the uncentered covariance, the second-moment matrix, of the input gradients of the loss: $\widetilde{G} = \frac{1}{n}\sum_i \nabla_x \ell_i\, \nabla_x \ell_i^\top$. Training minimizes $\mathcal{L}(B, a) = \frac{1}{2n}\sum_i r_i^2 + \frac{\lambda}{2}\|B\|_F^2$, so $\nabla_B \mathcal{L} = \frac{1}{n}QRX^\top + \lambda B$, and at exact B-stationarity

$$B = -\tfrac{1}{\lambda n}QRX^\top, \qquad \text{with defect} \qquad Z := \lambda n B + QRX^\top.$$

The residual weighting is not a modeling choice. It is the derivative of the square loss, and it enters B automatically.

3. The deterministic weighted square law

Let $T := QR^2Q^\top = n\widetilde{M}$, $S := QRX^\top XRQ^\top$, $d_{\text{eff}} := \langle S,T\rangle_F/\|T\|_F^2$ (the best Frobenius scalar fitting $S \approx dT$), and $\kappa_{\text{eff}} := d_{\text{eff}}/(\lambda^2 n)$. From $\lambda nB = -QRX^\top + Z$ we get $BB^\top = (\lambda^2n^2)^{-1}S + E_{\text{stat}}$ with $E_{\text{stat}} := (\lambda^2n^2)^{-1}(-QRX^\top Z^\top - ZXRQ^\top + ZZ^\top)$. Multiplying by $B^\top(\cdot)B$ and using $H^2 = B^\top(BB^\top)B$ gives the central identity.

Proposition 1 (Deterministic weighted law)**.** *With the notation above,*

$$H^2 - \kappa_{\text{eff}}\widetilde{G} = \tfrac{1}{\lambda^2n^2}\, B^\top(S - d_{\text{eff}}T)B + B^\top E_{\text{stat}}B, \qquad \textit{hence} \qquad \|H^2 - \kappa_{\text{eff}}\widetilde{G}\|_{\text{op}} \le \|B\|_{\text{op}}^2\left(\tfrac{\|T\|_{\text{op}}}{\lambda^2n^2}\,\mathcal{A}_{\text{pair}} + \|E_{\text{stat}}\|_{\text{op}}\right),$$

where $\mathcal{A}_{\text{pair}} := \left\|T^{\dagger/2}(S - d_{\text{eff}}T)T^{\dagger/2}\right\|_{\text{op}}$*. In particular, if $Z = 0$ and $S = d_{\text{eff}}T$ on* $\text{range}(T)$*, then $H^2 = \kappa_{\text{eff}}\widetilde{G}$ exactly.*

The pushed pair error is the operative quantity. The bound uses $\mathcal{A}_{\text{pair}}$, the worst pair-isotropy error over the whole support of T, but the law only sees the term $B^\top(S - d_{\text{eff}}T)B$. Let $QR = U\Sigma V^\top$ be the thin SVD ($U \in \mathbb{R}^{m\times r}$ in hidden space, $V \in \mathbb{R}^{n\times r}$ in sample space), and set $H_X := V^\top X^\top XV$, $F_X := H_X - d_{\text{eff}}I_r$, $G_{\text{stat}} := \Sigma^2V^\top X^\top$. Then $S - d_{\text{eff}}T = U\Sigma F_X\Sigma U^\top$, $\mathcal{A}_{\text{pair}} = \|F_X\|_{\text{op}}$, and at exact stationarity, since $B = -(\lambda n)^{-1}U\Sigma V^\top X^\top$,

$$B^\top(S - d_{\text{eff}}T)B = \tfrac{1}{\lambda^2n^2}\, G_{\text{stat}}^\top F_X\, G_{\text{stat}}.$$

So F_X is the pair-isotropy defect of the input geometry $X^\top X$ *restricted to the sample combinations the residual-weighted hidden gradients use*, G_{stat} weights each such direction by σ_k^2 and projects it through $X^\top$, and the weighted law breaks only when F_X is large on the high-gain directions of G_{stat}. A large global $\mathcal{A}_{\text{pair}} = \|F_X\|_{\text{op}}$ with all the defect on low-gain directions costs almost nothing. A sufficient probabilistic mechanism: if a high-gain right-singular subspace $W = VP_{\text{hi}}$ is fixed or independent of Gaussian X, then $W^\top X^\top XW \approx dI$ by standard Wishart concentration, so F_X is small there. The catch is that the learned W depends on X, so this is a sufficient condition, not a proof for trained networks (Sec. **??**). Upgrading it would require an adaptive-concentration argument over a low-complexity class of learnable subspaces.

4. From weighted to raw AGOP, and the failure taxonomy

Raw AGOP needs a scalar bridge $\widetilde{M} \approx \beta M$. The Frobenius best-fit one is $\beta_{\text{fit}} := \langle\widetilde{M}, M\rangle_F/\|M\|_F^2$. Using $\langle q_iq_i^\top, q_jq_j^\top\rangle_F = (q_i^\top q_j)^2$,

$$\beta_{\text{fit}} = \frac{\sum_{i,j} r_i^2(q_i^\top q_j)^2}{\sum_{i,j}(q_i^\top q_j)^2} = \frac{1}{n}\sum_i \ell_i\, r_i^2, \qquad \ell_i := \frac{\sum_j(q_i^\top q_j)^2}{\frac{1}{n}\sum_k\sum_j(q_k^\top q_j)^2}, \qquad \tfrac{1}{n}\sum_i \ell_i = 1,$$

i.e. β_{fit} is a hidden-gradient-leverage-weighted average of the squared residuals, with $\min_i r_i^2 \le \beta_{\text{fit}} \le \max_i r_i^2$. Hence

$$\frac{\beta_{\text{fit}}}{\bar{r}^2} - 1 = \text{Corr}_n(\ell, r^2)\ \text{CV}_n(\ell)\ \text{CV}_n(r^2), \qquad \bar{r}^2 := \tfrac{1}{n}\sum_i r_i^2,$$

so the beta bridge tracks mean residual energy precisely when hidden-gradient leverage and residual energy are weakly correlated. Combining with Prop. **??**: if $\|H^2 - \kappa_{\text{eff}}\widetilde{G}\|_{\text{op}} \le E_w$ and $\|\widetilde{G} - \beta_{\text{fit}}G\|_{\text{op}} \le E_\beta$, then $\|H^2 - \kappa_{\text{eff}}\beta_{\text{fit}}G\|_{\text{op}} \le E_w + \kappa_{\text{eff}}E_\beta$, and solving for G divides by $|\beta_{\text{fit}}|$. So the raw conversion is well conditioned only away from interpolation, where β_{fit} is bounded away from zero. This is the *two-regime picture*: $H^2 \approx \kappa_{\text{eff}}\widetilde{G}$ is the stable late object, while $H^2 \approx \kappa_{\text{eff}}\beta_{\text{fit}}G$ (raw-AGOP-like) is visible only in an intermediate regime.

Failure taxonomy. Three distinct mechanisms break NFA-type behavior. *(i) Beta failure*: $\text{Corr}_n(\ell, r^2)$ is large, so β_{fit} deviates from $\bar{r}^2$, the raw law is biased, but the weighted law is unaffected. *(ii) High-gain pair failure*: F_X is large on the high-gain directions of G_{stat}, so $G_{\text{stat}}^\top F_X G_{\text{stat}}$ is large and the weighted law itself fails. *(iii) Late-regime collapse*: near interpolation $\beta_{\text{fit}} \to 0$, so the raw conversion is ill-conditioned even when the weighted law is exact. A fourth case is benign: $\mathcal{A}_{\text{pair}}$ large but the pushed error small (the conservative-diagnostic case). The two links are logically independent: there are explicit 2-dimensional stationary instances ($n = 2$, $\lambda n = 1$) where (a) the weighted law is exact while $\widetilde{M} = \beta M$ has no scalar solution, (b) the beta bridge is exact while a high-gain pair defect breaks the weighted law, and (c) $\mathcal{A}_{\text{pair}}$ is order one while the pushed error is $O(\varepsilon^2)$. Consequently:

Corollary 1. *No theorem whose only hypotheses are exact B-stationarity and finite-dimensional algebra can imply a universal exact raw scalar NFA law $H \approx cG^\alpha$. The two bridge conditions (beta stability and high-gain pair closure) are required.*

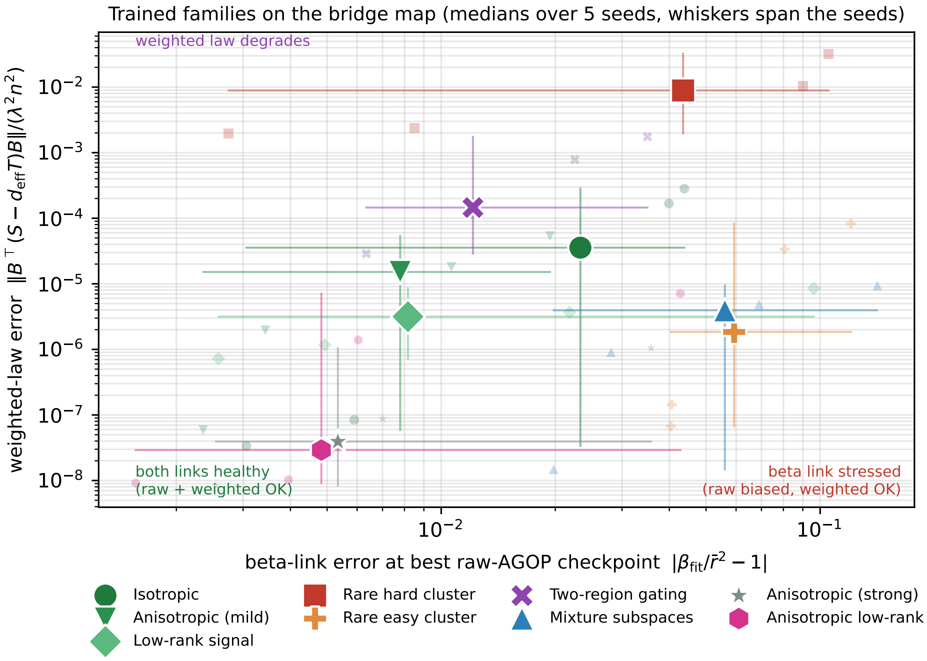

Figure 1: Trained families on the bridge map (5 seeds, markers at family medians, whiskers span the seeds). Axes: beta-link error $|\beta_{\mathrm{fit}}/\bar{r}^2 - 1|$ at the best raw-AGOP checkpoint (horizontal) versus weighted-law residual $\|B^\top(S - d_{\mathrm{eff}}T)B\|/(\lambda^2 n^2)$ at the best-stationarity checkpoint (vertical). The positive controls and the two pair-geometry attacks (strong anisotropy, anisotropic low-rank) sit in the bottom-left healthy cluster, the mixture and rare-easy families move right (beta link stressed, weighted law fine), and only the rare hard cluster moves up by orders of magnitude (two-region gating, mildly).

5. Experiments

Protocol. All data are synthetic teacher-student regression. For each run, a fixed random two-layer teacher $f_*(x) = a_*^\top \phi(B_* x)$ labels $n = 128$ inputs $x_i \in \mathbb{R}^{256}$ drawn from a chosen distribution (clean labels), and a fresh two-layer student ($m = 16$ hidden units) is trained on those pairs with square loss plus ridge on B ($\lambda = 10^{-2}$), using gradient descent followed by L-BFGS refinement, in double precision, logging the diagnostics at roughly 170 checkpoints. I run five random seeds per family. At each checkpoint I evaluate exactly the objects of Prop. **??** and Sec. 4: the weighted residual $\|H^2 - \kappa_{\mathrm{eff}}\widetilde{G}\|_{\mathrm{op}}$, the theorem bound, $\beta_{\mathrm{fit}}/\bar{r}^2$ and $\mathrm{Corr}_n(\ell, r^2)$, the global $\mathcal{A}_{\mathrm{pair}}$ versus the pushed pair error $\|B^\top(S - d_{\mathrm{eff}}T)B\|_{\mathrm{op}}/(\lambda^2 n^2)$, and the stationarity defect. Quantities are read at the best-stationarity checkpoint (smallest $\|BB^\top - B_{\mathrm{stat}}B_{\mathrm{stat}}^\top\|_{\mathrm{op}}$) unless noted. The data families are three positive controls (isotropic Gaussian, anisotropic Gaussian, low-rank signal plus isotropic noise) and six adversarial families built to attack a specific link (rare hard and rare easy high-leverage clusters, two-region gating, mixture of subspaces, strongly anisotropic Gaussian, and a low-rank signal with a steeply anisotropic within-subspace covariance).

Positive families. At best-stationarity checkpoints the weighted law is accurate and well within its deterministic bound: mean weighted residual $\approx 3.6 \times 10^{-3}$, 1.9×10^{-3}, 6.4×10^{-4} for the isotropic, anisotropic, and low-rank-signal families. The beta bridge is benign: $\beta_{\mathrm{fit}}/\bar{r}^2$ averages 0.99, 1.00, 1.02 across seeds, with small spread. And the global pair diagnostic is conservative: $\mathcal{A}_{\mathrm{pair}}$ averages roughly 80 to 280 while the pushed pair error averages $\approx 5.6 \times 10^{-5}$, 1.1×10^{-5}, 2.5×10^{-6}, five to seven orders of magnitude smaller, exactly as the SVD factorization predicts. Along training, the raw-AGOP error bottoms out in an intermediate window while the weighted-law error keeps decreasing into stationarity, and β_{fit} collapses with the mean residual energy near interpolation, so a single end-of-training correlation number hides the structure the decomposition makes explicit.

Trained bridge map and a robustness finding. Figure **??** places all nine families on axes given by the two links. The most surprising empirical observation is that the weighted law remains stable in more families than I expected. Two families built deliberately to attack high-gain pair closure, a strongly anisotropic Gaussian and a low-rank signal with a steeply anisotropic within-subspace covariance, land in the healthy bottom-left cluster with the positive controls, with pushed pair errors near 10^{-7} despite sizeable global $\mathcal{A}_{\mathrm{pair}}$ (and high-gain closure norms). The reason looks structural: near stationarity $B = -(\lambda n)^{-1}QRX^\top$, so the pushed error only sees the directions the residual-weighted hidden gradients span, and the network appears to self-align those with the input geometry. The mixture-of-subspaces and rare-easy families instead move right: they stress the beta link ($|\beta_{\rm fit}/\bar{r}^2 - 1| \approx 0.05$, raw AGOP biased), while the weighted law stays fine. The only family that moves up by orders of magnitude is the rare hard cluster (and, mildly, two-region gating): there, amplified outlier labels keep a small set of residuals large persistently, sustaining a misaligned QR the network cannot iron out at stationarity, and that same persistence also bends the beta link, so it is a combined failure rather than a pure pair one. In these runs the weighted law degrades only under persistent-residual outlier structure.

6. Discussion

Relation to recent work. AGOP and recursive feature machines [?] make AGOP a central object of feature learning, and FACT [?] derives convergence-side feature relations from first-order optimality and explains when NFA holds. The present work is the concrete two-layer square-loss specialization of that mechanism, with the addition being the bridge decomposition: identifying $\widetilde{G}$ as the loss-gradient second moment, isolating the beta and high-gain-pair links, showing the global pair diagnostic is conservative, and mapping which link fails where. Deep-NFA theory proves NFA in linear regimes with nonlinear counterexamples [?], consistent with Corollary **??**. Concurrent work studies $B^\top B$ as a feature-dynamics object via a Virtual Covariance and a Feature Learning Equation [?], and relating that quantity to $\widetilde{G}$ is a natural next step. Feature-learning dynamics from small initialization is treated by alternating gradient flows [?], a candidate tool for the open dynamics direction below.

Limitations. The model is two-layer with square loss and ridge on B only, and the families are synthetic. The deterministic identity assumes exact stationarity. I report near it and log $E_{\rm stat}$ so its contribution is visible, and the theorem bound uses the conservative $\mathcal{A}_{\rm pair}$, so the bound ratio is well below one rather than tight. The high-gain pair-closure result is conditional (a frozen-subspace concentration mechanism), not a proof for trained networks. The trained map is a guiding map across nine families, not a deliberate grid sweep over the leverage-residual correlation crossed with high-gain anisotropy, and per-seed spread is wide while family medians separate cleanly.

Open directions. (i) Derive the beta decorrelation (or high-gain pair closure) from training dynamics rather than assuming it. The relevant empirical probe, whether high hidden-gradient leverage predicts faster residual decay, is logged. (ii) Isolate a pure high-gain pair failure: every pair-geometry attack so far self-aligns near stationarity. (iii) A full deliberate phase diagram with samplewise diagnostics, and one non-synthetic dataset. (iv) Extend the law beyond two layers by replacing B and q_i with layerwise Jacobian factors.

7. Conclusion

In a two-layer model trained with square loss and ridge on the first layer, first-order stationarity does not predict raw AGOP. It predicts a residual-weighted AGOP law $H^2 \approx \kappa_{\rm eff}\widetilde{G}$, with $\widetilde{G}$ equal to the uncentered second moment of the input gradients of the loss, and with an exact error split into a pushed pair-isotropy term and a stationarity defect. Raw AGOP is a derived, regime-dependent consequence, gated by a beta bridge and high-gain pair closure, which are logically independent and individually checkable, so no universal raw NFA law follows from stationarity alone. Empirically, the weighted law is robust: across synthetic families it degrades only under persistent-residual outlier structure, and even that case is a combined beta-and-pair failure rather than a pure pair one. The takeaway is that AGOP/NFA-type relations are governed by beta conditioning and high-gain pair closure, and the residual-weighted law is the stable convergence-side object.

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